CITI × NPCI DRUNIX HACKATHON

REAL-TIME PAYMENTS / STAGE 2

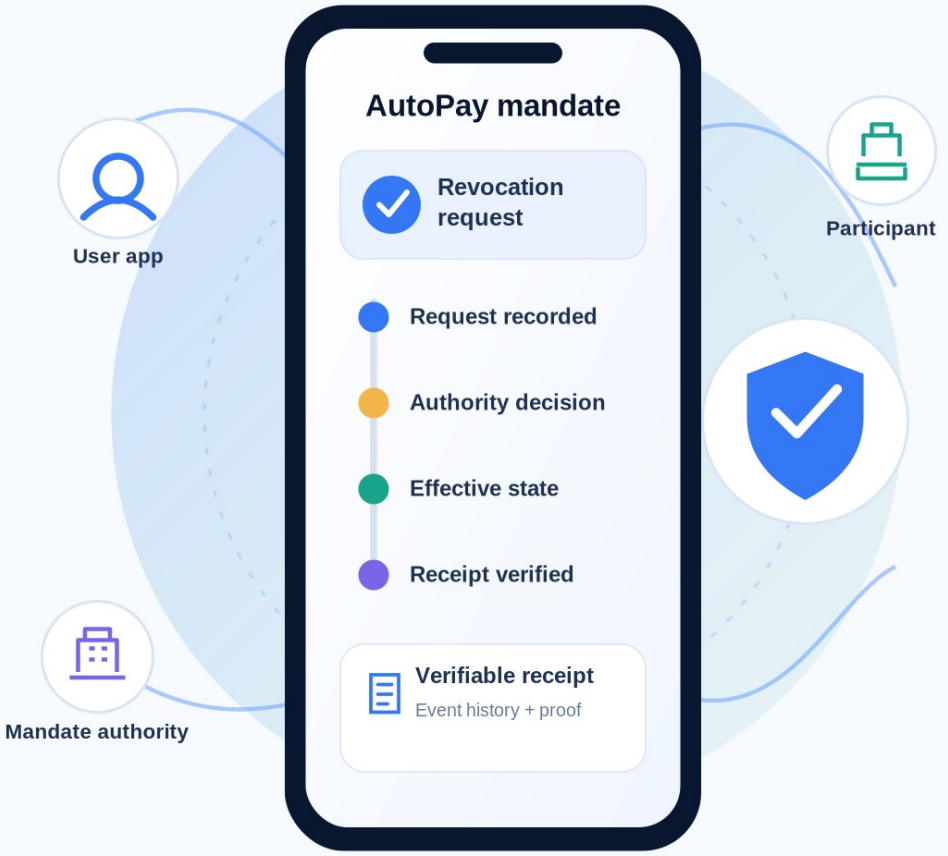
MandateGuard

Cross-Participant Proof of UPI AutoPay Revocation

A DRUNIX-based layer to record, reconcile and verify mandate-revocation events across participating systems.

01	02	03
REQUEST	CONFIRM	VERIFY

Concept proposal • Simulated MVP • 12-day build plan



The ledger proves the recorded sequence—not the truth of every input.

A revocation request is not the same as an effective revocation.

The opportunity is verifiable coordination across participants—not replacing existing mandate controls.

User / UPI app

Needs a clear record distinguishing request submission from confirmed effective status.

Mandate-handling participant

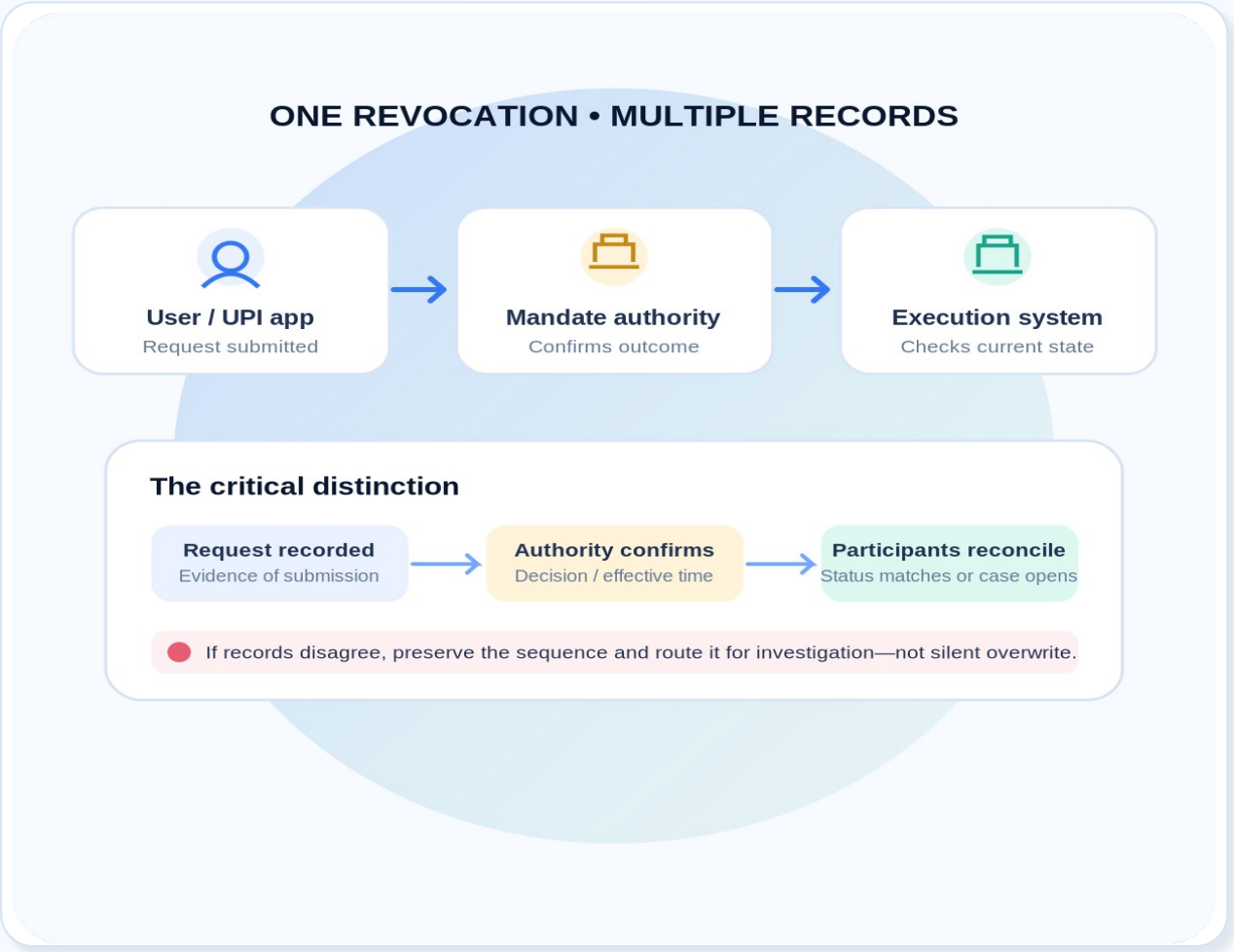
Must remain authoritative for the revocation decision and effective time.

Payment executor / merchant

Needs the latest applicable mandate status in any integrated debit workflow.

What MandateGuard targets

A traceable event sequence and a route to resolve missing or conflicting acknowledgements.



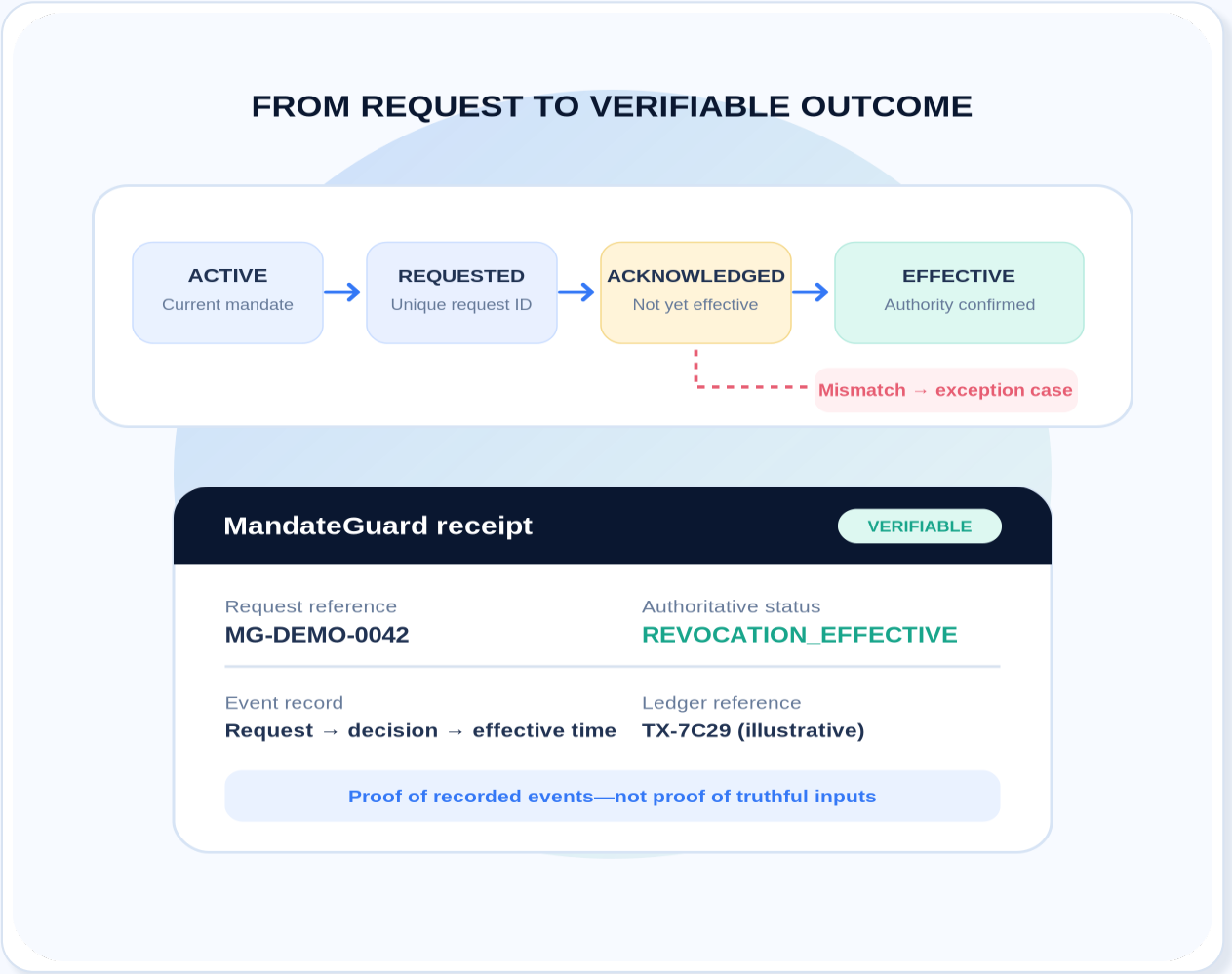
Policy context: NPCI UPI OC-223/2025-26 (7 Oct 2025) already includes mandate lifecycle management.

Separate the request, acknowledgement and effective state.

MandateGuard makes each transition explicit, attributable and independently checkable.

1. **Capture request**
   Create a unique request reference and idempotency key.
2. **Record authority outcome**
   Only the authorized mandate-handling participant confirms the decision.
3. **Reconcile participant state**
   Missing or conflicting acknowledgements open an exception; they do not silently overwrite history.
4. **Issue verifiable receipt**
   Link event order, timestamps, participant identity and ledger reference.
5. **Check before debit (simulated)**
   The demo executor checks the effective state before a scheduled-debit scenario.

Guardrail: a submitted request is never labelled “effective” by default.



DRUNIX is the shared state layer—not a blockchain sticker.

The core contract controls lifecycle transitions; the UI and integrations support that governed workflow.



Reference: NPCI DRUNIX architecture guide — endorsement/commit separation, state DB options and KeyDB transient store.

Core components

Go chaincode

Mandate state machine, request uniqueness, authorized transitions and exception events.

Multi-organisation endorsement

Critical updates require the configured authorized participants, not a UI toggle.

CouchDB + restricted private data

CouchDB is the proposed queryable state store; sensitive payloads stay off-ledger or restricted.

Trust boundary

Only an integrated payment executor can enforce a live pre-debit check. The MVP simulates this boundary.

A focused 12-day MVP. Measurable, not hand-waved.

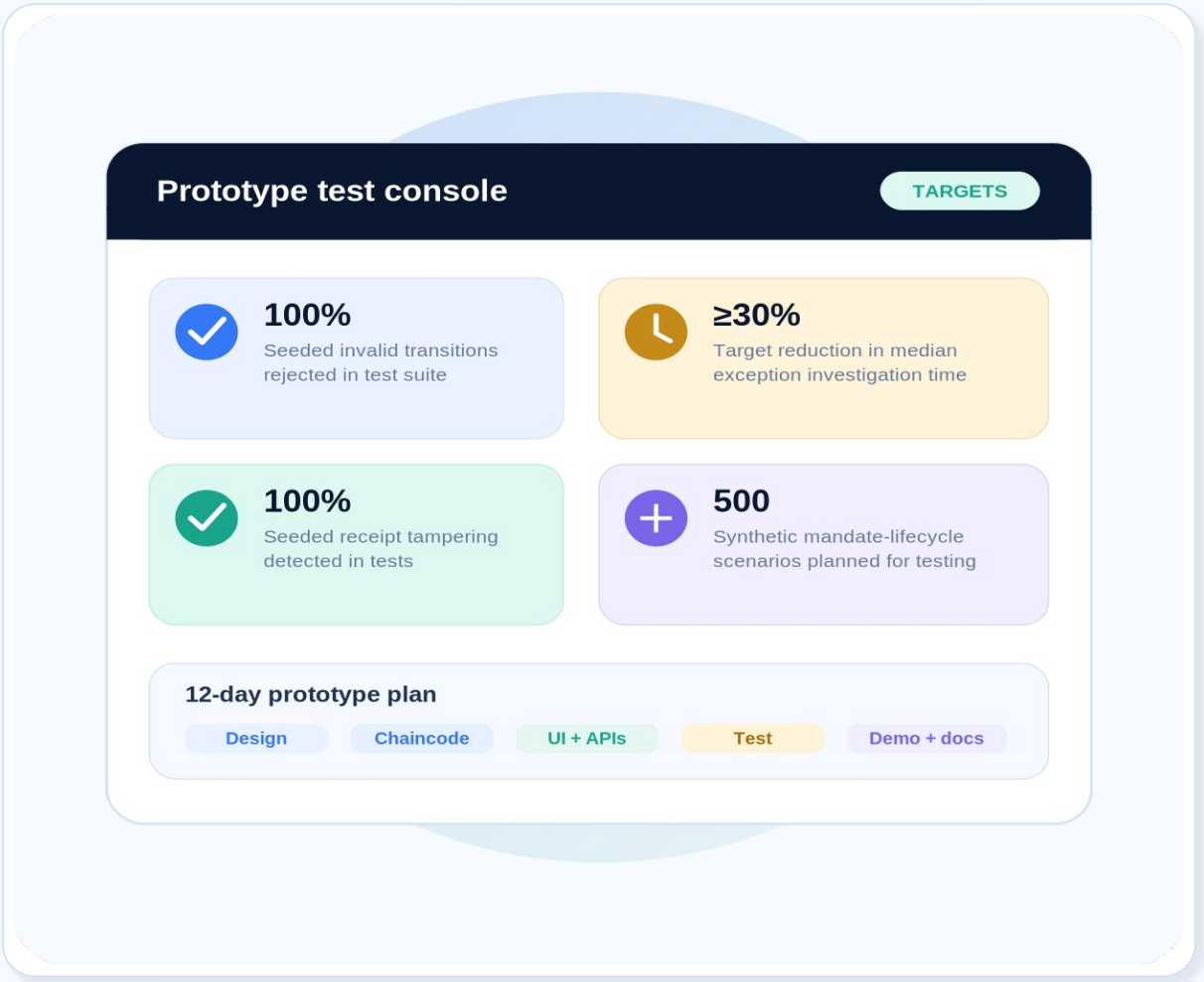
All figures below are proposed targets to validate in a simulated environment—not achieved results.

What the judge can test

A	**Normal revocation**	Request → authority decision → effective state → receipt
B	**Delayed acknowledgement**	Exception opens; state is not falsely marked effective
C	**Replay / invalid transition**	Duplicate request and stale transition are rejected
D	**Tampered receipt**	Verification flags changed receipt fields
E	**Post-revocation debit attempt**	Simulated executor rejects the debit scenario

12 DAYS • ₹2,000 ESTIMATED PROTOTYPE BUDGET

React · Node.js/Express · Go chaincode · DRUNIX · synthetic data



Dependencies: DRUNIX dev/test network + SDK; organizer guidance on authority boundary; payment sandbox optional.